Chi-Square Tests & Regression Inference

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Chi-square goodness-of-fit test**

A die is rolled 60 times, expecting each face to appear equally often (expected count 10 per face). The observed counts are $12, 8, 11, 9, 7, 13$. Compute the chi-square test statistic using $\chi^2 = \sum \frac{(O-E)^2}{E}$.

$$\chi^2 = \frac{(12-10)^2}{10} + \frac{(8-10)^2}{10} + \frac{(11-10)^2}{10} + \frac{(9-10)^2}{10} + \frac{(7-10)^2}{10} + \frac{(13-10)^2}{10} = \frac{4+4+1+1+9+9}{10} = \frac{28}{10} = 2.8.$$

2 **[Foundational]** • **Target: Chi-square goodness-of-fit test**

State the null and alternative hypotheses for the goodness-of-fit test in the die-rolling scenario above.

H_0: the die is fair (each face has probability $\frac{1}{6}$). H_a: the die is not fair (at least one face's true probability differs from $\frac{1}{6}$).

3 **[Foundational]** • **Target: Chi-square test for independence**

A two-way table examines whether preferred learning style (visual, auditory, kinesthetic) is associated with academic major (STEM, non-STEM). State the null and alternative hypotheses for a chi-square test of independence in this context.

H_0: preferred learning style and academic major are independent (no association). H_a: preferred learning style and academic major are not independent (there is an association).

4 **[Foundational]** • **Target: Chi-square test for independence**

In a chi-square test of independence with a 3×2 table (3 learning styles, 2 major categories), find the degrees of freedom, using $df = (\text{rows} - 1)(\text{columns} - 1)$.

$df = (3-1)(2-1) = 2 \times 1 = 2$.

5 **[Foundational]** • **Target: Interpreting chi-square results**

A chi-square test of independence produces a test statistic $\chi^2 = 8.2$ with $df = 2$, and a p-value of 0.017. State a conclusion at the $\alpha = 0.05$ significance level, in context.

Since the p-value (0.017) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence of an association between the two categorical variables being studied.

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6 [Foundational] • Target: Interpreting chi-square results

Explain a condition that must be checked before conducting a chi-square test, related to expected cell counts.

All expected counts in the table should be at least 5 (a common guideline). If some expected counts are too small, the chi-square approximation may not be reliable, and results should be interpreted with caution or a different method used.

7 [Foundational] • Target: Inference for the slope of a regression line

A regression of exam score on hours studied gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.1$, based on $n = 25$ data points. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.069$ (with $df = 23$).

Margin of error: $2.069(1.1) \approx 2.28$. Confidence interval: 6.2 ± 2.28, or approximately $(3.92, 8.48)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Inference for the slope of a regression line

State the null and alternative hypotheses for testing whether hours studied has a significant linear relationship with exam score, and compute the test statistic using the values from the previous problem.

$H_0 : \beta_1 = 0$ (no linear relationship). $H_a : \beta_1 \neq 0$ (there is a linear relationship). Test statistic: $t = \frac{b_1 - 0}{SE_{b_1}} = \frac{6.2}{1.1} \approx 5.64$.

9 [Intermediate] • Target: Conditions for regression inference

State the four conditions (sometimes remembered as LINER: Linear, Independent, Normal, Equal variance, Random) required for valid inference about a regression slope.

(1) Linear: the true relationship between x and y is linear. (2) Independent: individual observations are independent of each other. (3) Normal: for any fixed x-value, the responses y vary according to a normal distribution. (4) Equal variance: the standard deviation of y is the same for all values of x (no fanning pattern in the residuals). Additionally, data should come from a properly randomized design or random sample.

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10 **[Intermediate]** • **Target: Putting it together: a full inference scenario**

A study finds a chi-square test statistic of $\chi^2 = 3.1$ with $df = 1$, and a critical value of $\chi^2_{0.05} = 3.84$ at the $\alpha = 0.05$ level. State the conclusion, and explain how this differs from a p-value based approach.

Since the test statistic (3.1) is less than the critical value (3.84), we fail to reject H_0 — there is not enough evidence of an association. Using critical values compares the test statistic directly to a threshold on the chi-square distribution, while a p-value approach compares a computed probability to α directly; both lead to the same reject/fail-to-reject decision when done correctly, since the critical value is simply the test-statistic value corresponding to a p-value of exactly α.

11 **[Intermediate]** • **Target: Chi-square test for homogeneity**

Explain the difference between a chi-square test of independence and a chi-square test of homogeneity, even though both use the same formula and table structure.

A test of independence uses a single sample and asks whether two categorical variables measured on the same individuals are associated. A test of homogeneity compares the distribution of a single categorical variable across two or more separate populations or groups (e.g. comparing the distribution of preferred learning style between two different schools) — the calculation and table setup are identical, but the sampling design and the question being asked differ.

12 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

In a two-way table, a row total is 80, a column total is 120, and the grand total is 300. Find the expected count for that cell, using $E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$.

$E = \frac{(80)(120)}{300} = \frac{9600}{300} = 32.$

13 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

State the degrees of freedom for a chi-square goodness-of-fit test with 6 categories, using $df = (\text{number of categories}) - 1$.

$df = 6 - 1 = 5.$

14 **[Intermediate]** • **Target: Computing expected counts for a chi-square test**

A genetics experiment expects offspring to appear in a 9:3:3:1 ratio across four phenotype categories, based on 160 total offspring observed. Find the expected count for each category, and briefly describe what the chi-square goodness-of-fit test would examine using these expected counts.

Total ratio parts: $9 + 3 + 3 + 1 = 16$. Expected counts: $\frac{9}{16}(160) = 90$, $\frac{3}{16}(160) = 30$, $\frac{3}{16}(160) = 30$, $\frac{1}{16}(160) = 10$. The chi-square goodness-of-fit test would compare these expected counts to the actual observed counts in each category, computing $\chi^2 = \sum \frac{(O-E)^2}{E}$ to assess whether the observed offspring distribution is consistent with the predicted 9:3:3:1 genetic ratio.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Computing expected counts for a chi-square test** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A study examines the relationship between hours of sleep per night (x) and reaction time in milliseconds (y) for 30 participants, finding a regression slope $b_1 = -8.5$ with standard error $SE_{b_1} = 2.3$. State the hypotheses for testing whether more sleep is associated with a significant decrease in reaction time (a one-sided test), compute the test statistic, and interpret the sign of the slope in context.

$H_0 : \beta_1 = 0$ (no linear relationship). $H_a : \beta_1 < 0$ (more sleep is associated with decreasing reaction time). Test statistic: $t = \frac{b_1 - 0}{SE_{b_1}} = \frac{-8.5}{2.3} \approx -3.70$. The negative slope indicates that, in this sample, reaction time tends to decrease (improve) as hours of sleep increase — consistent with the idea that more sleep is associated with faster reaction times.

16 **[Challenging]** • **Target: Computing expected counts for a chi-square test** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A university surveys 240 students, recording both their preferred study location (library, dorm, coffee shop) and their class year (underclassman, upperclassman), to test whether study location preference is independent of class year. The resulting chi-square test statistic is $\chi^2 = 11.4$ with $df = 2$, and the corresponding p-value is 0.0033. State the conclusion at $\alpha = 0.01$, and explain, in the context of this scenario, what rejecting or failing to reject H_0 would mean for the university's understanding of its students.

Since the p-value (0.0033) is less than $\alpha = 0.01$, we reject H_0. This means there is convincing statistical evidence of an association between preferred study location and class year — in context, this suggests underclassmen and upperclassmen tend to have systematically different study location preferences, which could inform decisions like where the university invests in study space improvements for each group.

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17 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher studies the relationship between advertising spending (in thousands of dollars, x) and monthly sales (in thousands of dollars, y) for 22 small businesses, obtaining a regression slope $b_1 = 3.2$ with standard error $SE_{b_1} = 0.85$. Construct a 95% confidence interval for the true population slope β_1 using a critical value $t^* \approx 2.086$ (with $df = 20$), and interpret the interval in context, including whether it suggests advertising spending has a statistically significant positive association with sales.

Margin of error: $2.086(0.85) \approx 1.773$. Confidence interval: 3.2 ± 1.773, or approximately $(1.427, 4.973)$. Interpretation: we are 95% confident that, for every additional thousand dollars spent on advertising, monthly sales increase by somewhere between about 1.427 and 4.973 thousand dollars, on average. Since this entire interval is positive (it does not contain 0), this suggests a statistically significant positive association between advertising spending and sales.

18 [Challenging] • Target: Putting it together: a full inference scenario • Cross-Lesson: 3.03 The Normal Distribution

Eight patients with chronic insomnia try a new relaxation therapy. Their sleep duration (hours/night) is measured before and after 4 weeks of therapy: Patient differences $d = \text{After} - \text{Before}$ yield $\bar{x}_d = 1.25$ hours and $s_d = 0.80$ hours. Assuming normality of differences, test whether the therapy significantly increased sleep at $\alpha = 0.01$.

1. Method Choice: Paired data measured on the same 8 patients $\Rightarrow$ Paired t-test on differences. 2. Hypotheses: $H_0 : \mu_d = 0 \quad \text{vs.} \quad H_a : \mu_d > 0 \quad (\alpha = 0.01),$ where μ_d is the true mean difference in sleep duration (After minus Before). 3. Test Statistic & p-value: $t = \frac{\bar{x}_d - 0}{s_d/\sqrt{n}} = \frac{1.25 - 0}{0.80/\sqrt{8}} = \frac{1.25}{0.2828} \approx 4.42 \quad (df = 8 - 1 = 7).$ With $df = 7$, $P(T > 4.42) \approx 0.0015$. 4. Conclusion: Because $\text{p} - \text{value} = 0.0015 < \alpha = 0.01$, we reject H_0. There is convincing statistical evidence that the relaxation therapy significantly increased sleep duration in patients.

19 [Challenging] • Target: Chi-square test for homogeneity • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In the tomato experiment in Problem 7, the agronomist has two distinct tomato varieties: Roma and Beefsteak. (a) Explain why a randomized block design is superior to a completely randomized design. (b) Outline the blocking protocol.

(a) Superiority of Blocking: Tomato variety is known to have a massive effect on fruit yield. A completely randomized design risks assigning disproportionately more Beefsteak plants to one fertilizer, confounding variety with fertilizer efficacy. Blocking by variety isolates and eliminates the variability between varieties from the experimental error, increasing statistical power. (b) Blocking Protocol: Separate the plants into two blocks: Block 1 (Roma) and Block 2 (Beefsteak). Within Block 1, randomly assign half the Roma plants to organic fertilizer and half to standard fertilizer. Within Block 2, randomly assign half the Beefsteak plants to organic fertilizer and half to standard fertilizer independently. Compare yields between treatments separately within each block.

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20 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A researcher is investigating the relationship between study habits and exam performance. They categorize students into two groups: 'Regular Study' and 'Cramming'. They also categorize exam performance into 'Pass' and 'Fail'. The data collected is presented in a two-way table. If the table has 2 rows representing study habits and 3 columns representing exam performance categories, what is the total number of cells in this two-way table?

A two-way table displays the frequencies of two categorical variables. The number of rows represents the number of categories for the first variable, and the number of columns represents the number of categories for the second variable. To find the total number of cells in the table, we multiply the number of rows by the number of columns. Number of rows = 2 Number of columns = 3 Total number of cells = Number of rows $\times$ Number of columns = $2 \times 3 = 6$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Chi-square goodness-of-fit test

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $17, 21, 13, 9$. Compute the chi-square test statistic using $\chi^2 = \sum (O - E)^2/E$.

a. 5.33 ✓

b. 4.33

c. 5.00

d. 6.33

$\chi^2 = (4 + 36 + 4 + 36)/15 = 80/15 \approx 5.33$.

22 [Foundational] • Target: Chi-square test for independence

A survey expects 4 categories to appear equally often, with expected count $E = 20$ per category. The observed counts are $24, 14, 18, 24$. Compute the chi-square test statistic using $\chi^2 = \sum (O - E)^2/E$.

a. 4.60

b. 3.60 ✓

c. 2.60

d. 3.43

$\chi^2 = (16 + 36 + 4 + 16)/20 = 72/20 \approx 3.60$.

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23 [Foundational] • Target: Interpreting chi-square results

A survey expects 4 categories to appear equally often, with expected count $E = 11$ per category. The observed counts are $6, 16, 9, 13$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 4.83

b. 6.27

c. 4.27

d. 5.27 ✓

$\chi^2 = (25 + 25 + 4 + 4)/11 = 58/11 \approx 5.27$.

24 [Foundational] • Target: Inference for the slope of a regression line

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $10, 18, 12, 20$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 5.53

b. 4.53 ✓

c. 4.25

d. 3.53

$\chi^2 = (25 + 9 + 9 + 25)/15 = 68/15 \approx 4.53$.

25 [Foundational] • Target: Conditions for regression inference

A survey expects 4 categories to appear equally often, with expected count $E = 24$ per category. The observed counts are $27, 26, 25, 18$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 3.08

b. 2.00

c. 2.08 ✓

d. 1.08

$\chi^2 = (9 + 4 + 1 + 36)/24 = 50/24 \approx 2.08$.

26 [Foundational] • Target: Putting it together: a full inference scenario

A survey expects 4 categories to appear equally often, with expected count $E = 15$ per category. The observed counts are $21, 9, 16, 14$. Compute the chi-square test statistic using $\chi^2 = \sum(O - E)^2/E$.

a. 3.93

b. 4.93 ✓

c. 5.93

d. 4.62

$\chi^2 = (36 + 36 + 1 + 1)/15 = 74/15 \approx 4.93$.

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27 [Foundational] • Target: Chi-square test for homogeneity

A survey expects 4 categories to appear equally often, with expected count $E = 14$ per category. The observed counts are $15, 18, 11, 12$. Compute the chi-square test statistic using $\chi^2 = \sum (O - E)^2 / E$.

a. 1.14

b. 3.14

c. 2.00

d. 2.14 ✓

$\chi^2 = (1 + 16 + 9 + 4)/14 = 30/14 \approx 2.14$.

28 [Foundational] • Target: Computing expected counts for a chi-square test

In a two-way table, a row total is 87, a column total is 46, and the grand total is 200. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 23.01

b. 112.99

c. 20.01 ✓

d. 17.01

$E = (87)(46)/200 = 4002/200 \approx 20.01$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Chi-Square Goodness-of-Fit Test

In a two-way table, a row total is 266, a column total is 99, and the grand total is 400. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 68.83

b. 65.83 ✓

c. 62.84

d. 299.17

$E = (266)(99)/400 = 26334/400 \approx 65.83$.

30 [Intermediate] • Target: Chi-Square Test of Independence vs. Homogeneity

In a two-way table, a row total is 250, a column total is 314, and the grand total is 500. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 160.00

b. 157.00 ✓

c. 407.00

d. 154.00

$E = (250)(314)/500 = 78500/500 \approx 157.00$.

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31 [Intermediate] • Target: Inference for Linear Regression Slope

In a two-way table, a row total is 103, a column total is 50, and the grand total is 150. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 118.67

b. 31.33

c. 37.33

d. 34.33 ✓

$E = (103)(50)/150 = 5150/150 \approx 34.33$.

32 [Intermediate] • Target: Regression Model Conditions (LINER)

In a two-way table, a row total is 154, a column total is 295, and the grand total is 500. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 358.14

b. 87.86

c. 90.86 ✓

d. 93.86

$E = (154)(295)/500 = 45430/500 \approx 90.86$.

33 [Intermediate] • Target: Chi-square goodness-of-fit test

In a two-way table, a row total is 96, a column total is 134, and the grand total is 200. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 64.32 ✓

b. 61.32

c. 165.68

d. 67.32

$E = (96)(134)/200 = 12864/200 \approx 64.32$.

34 [Intermediate] • Target: Chi-square test for independence

In a two-way table, a row total is 65, a column total is 80, and the grand total is 250. Find the expected count for that cell, using $E =$ row total $\times$ column total / grand total.

a. 23.80

b. 17.80

c. 124.20

d. 20.80 ✓

$E = (65)(80)/250 = 5200/250 \approx 20.80$.

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35 [Intermediate] • Target: Interpreting chi-square results

A regression gives a sample slope $b_1 = -5.2$ with standard error $SE_{b_1} = 1.4$, based on data with $df = 33$. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.03$.

a. $(-8.04, -2.86)$

b. $(-9.04, -1.36)$

c. $(-7.54, -2.36)$

d. $(-8.04, -2.36)$ ✓

Margin of error: $2.03(1.4) \approx 2.84$. Interval: -5.2 ± 2.84, or approximately $(-8.04, -2.36)$.

36 [Intermediate] • Target: Inference for the slope of a regression line

A regression gives a sample slope $b_1 = 3.1$ with standard error $SE_{b_1} = 0.6$, based on data with $df = 15$. Construct a 95% confidence interval for the true population slope β_1, using a critical value $t^* \approx 2.131$.

a. $(1.82, 3.88)$

b. $(0.82, 5.38)$

c. $(2.32, 4.38)$

d. $(1.82, 4.38)$ ✓

Margin of error: $2.131(0.6) \approx 1.28$. Interval: 3.1 ± 1.28, or approximately $(1.82, 4.38)$.

37 [Intermediate] • Target: Conditions for regression inference

A regression gives a sample slope $b_1 = 6.8$ with standard error $SE_{b_1} = 1.25$. Compute the test statistic for testing $H_0 : \beta_1 = 0$, using $t = b_1/SE_{b_1}$, to two decimal places.

a. 6.44

b. -5.44

c. 4.44

d. 5.44 ✓

$t = b_1/SE_{b_1} = 6.8/1.25 \approx 5.44$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Putting it together: a full inference scenario • Cross-Lesson: 3.03 The Normal Distribution

A regression gives a sample slope $b_1 = 6.2$ with standard error $SE_{b_1} = 1.25$. Compute the test statistic for testing $H_0 : \beta_1 = 0$, using $t = b_1/SE_{b_1}$, to two decimal places.

a. -4.96

b. 3.96

c. 4.96 ✓

d. 5.96

$t = b_1/SE_{b_1} = 6.2/1.25 \approx 4.96$.

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39 [Challenging] • Target: Chi-square test for homogeneity • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A chi-square test produces a test statistic $\chi^2 = 13.7$ with $df = 5$, and a critical value of $\chi^2_{0.05} = 11.07$ at $\alpha = 0.05$. State the conclusion, comparing the test statistic directly to the critical value.

a. The test cannot be completed without a p-value

b. **Reject H_0; there is convincing evidence of an association** ✓

c. Fail to reject H_0; there is not convincing evidence of an association

d. Accept H_0 as definitely true

Since the test statistic (13.7) is greater than the critical value (11.07), we reject H_0. Reject H_0; there is convincing evidence of an association.

40 [Challenging] • Target: Computing expected counts for a chi-square test • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A chi-square test produces a test statistic $\chi^2 = 1.2$ with $df = 6$, and a critical value of $\chi^2_{0.05} = 12.59$ at $\alpha = 0.05$. State the conclusion, comparing the test statistic directly to the critical value.

a. The test cannot be completed without a p-value

b. Accept H_0 as definitely true

c. **Fail to reject H_0; there is not convincing evidence of an association** ✓

d. Reject H_0; there is convincing evidence of an association

Since the test statistic (1.2) is less than the critical value (12.59), we fail to reject H_0. Fail to reject H_0; there is not convincing evidence of an association.

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41 [Challenging] • Target: Chi-Square Goodness-of-Fit Test • Cross-Lesson: 1.01 Measures of Center & Spread

What is the formula for the Chi-Square test statistic?

a. $\chi^2 = \sum \frac{(O-E)^2}{E}$ ✓

b. $\chi^2 = \sum \frac{(O-E)}{E}$

c. $\chi^2 = \sum \frac{(O-E)^2}{O}$

d. $\chi^2 = \frac{(\bar{x}-\mu)^2}{s^2}$

The Chi-square statistic measures discrepancy between observed counts O and expected counts E: $\chi^2 = \sum \frac{(O-E)^2}{E}$.

42 [Challenging] • Target: Chi-Square Goodness-of-Fit Test • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

What condition must be satisfied regarding expected counts for a Chi-Square test to be valid?

a. **All expected cell counts must be at least** 5 ✓

b. All observed cell counts must be at least 30

c. The sample size must be exactly 100

d. The categories must be normally distributed

The standard Large Counts condition for Chi-square tests requires all expected counts to be at least 5.

43 [Challenging] • Target: Chi-Square Test of Independence vs. Homogeneity • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

What are the degrees of freedom for a Chi-Square test on a 3×4 contingency table?

a. 6 ✓

b. 12

c. 7

d. 11

$df = (r-1)(c-1) = (3-1)(4-1) = 2 \times 3 = 6$.

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44 [Challenging] • Target: Inference for Linear Regression Slope • Cross-Lesson: 2.01 Sampling Methods & Bias

Computer regression output shows: Predictor Coef SE Coef T P; Constant 14.2 2.1 6.76 0.000; Slope 3.45 0.69 5.00 0.0001. For $n = 18$ observations, what is the t-statistic for the slope?

a. 5.00 ✓

b. 6.76

c. 3.45

d. 0.69

The t-statistic for the slope is $t = \frac{b - 0}{SE_b} = \frac{3.45}{0.69} = 5.00$.

45 [Challenging] • Target: Regression Model Conditions (LINER) • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

What does the acronym LINER stand for regarding the conditions for regression inference?

a. **Linear, Independent, Normal y for each x, Equal variance, Random sampling** ✓

b. Least squares, Intercept, Normal, Errors, Residuals

c. Linear, Integrated, Numerical, Estimated, Robust

d. Limit, Interval, Normal, Expected, Ratio

The regression conditions are: Linear relationship, Independent observations, Normal distribution of response y at each x, Equal variance of residuals (homoscedasticity), and Random data collection.

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